Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2024 – Paper 2 – Solutions**

June 2024 – Question 1 – Solution

(a) Given that $f(x) = \frac{4}{x-5}$ and $g(x) = \frac{1}{x}$, determine the domain of the composite function $f\big(g(x)\big)$. [5]

$$f\big(g(x)\big) = f\left(\frac{1}{x}\right)$$

$$= \frac{4}{\frac{1}{x} - 5}$$

$$= 4 \div \left(\frac{1}{x} - 5\right)$$

$$= 4 \div \frac{1-5x}{x}$$

$$= 4 \times \frac{x}{1-5x}$$

$$= \frac{4x}{1-5x}$$

A function is discontinuous when the denominator is equal to zero.

$$1 - 5x = 0$$

$$5x = 1$$

$$x = \frac{1}{5}$$

$\therefore$ The domain of the composite function $f\big(g(x)\big)$ is $x \in \mathbb{R}, x \neq \frac{1}{5}$.

(b) Solve the inequality $-3|2x-5|+2 \geq -4$. [4]

$-3|2x-5|+2 \geq -4$

$-3|2x-5| \geq -4-2$

$-3|2x-5| \geq -6$

$|2x-5| \leq \frac{-6}{-3}$

$|2x-5| \leq 2$

<u>Case 1:</u>

$-(2x-5) \leq 2$

$2x-5 \geq \frac{2}{-1}$

$2x-5 \geq -2$

$2x \geq -2+5$

$2x \geq 3$

$x \geq \frac{3}{2}$

<u>Case 2:</u>

$(2x-5) \leq 2$

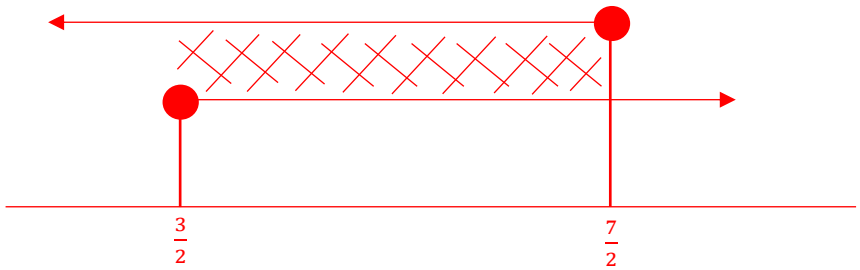
$2x \leq 2+5$

$2x \leq 7$

$x \leq \frac{7}{2}$

So, we have $x \geq \frac{3}{2}$ and $x \leq \frac{7}{2}$.

Consider,



$\therefore$ Solution set is $\left\{x: \frac{3}{2} \leq x \leq \frac{7}{2}\right\}$.

<u>Alternative Method:</u>

We have $|2x - 5| \leq 2$.

Use the algebraic property: $|x|^2 = x^2$.

Squaring both sides:

$$(2x - 5)^2 \leq (2)^2$$

$$4x^2 - 20x + 25 \leq 4$$

$$4x^2 - 20x + 25 - 4 \leq 0$$

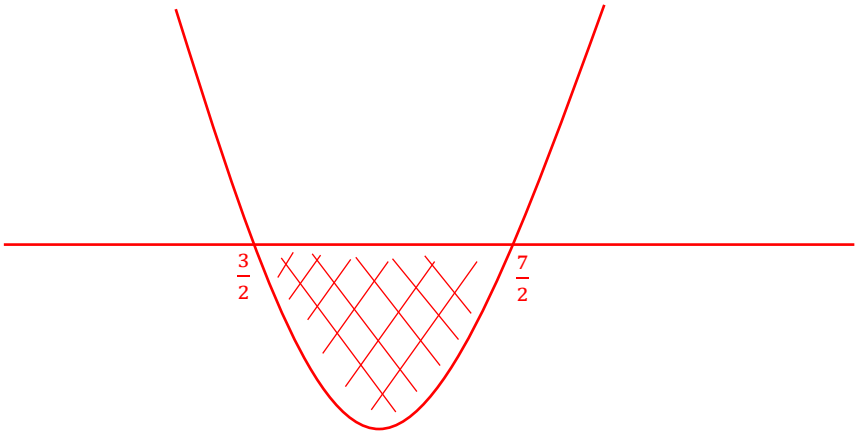
$$4x^2 - 20x + 21 \leq 0$$

$$4x^2 - 14x - 6x + 21 \leq 0$$

$$2x(2x - 7) - 3(2x - 7) \leq 0$$

$$(2x - 3)(2x - 7) \leq 0$$

The critical values are $\frac{3}{2}$ and $\frac{7}{2}$.



We are looking at the area where $(2x-3)(2x-7) \leq 0$.

$\therefore$ Solution set is $\left\{x: \frac{3}{2} \leq x \leq \frac{7}{2}\right\}$.

(c) Solve the logarithmic equation $\log_5 x - 4\log_x 5 = -3$. [8]

$$\log_5 x - 4\log_x 5 = -3$$

$$\log_5 x - 4\left(\frac{\log_5 5}{\log_5 x}\right) = -3$$

$$\log_5 x - 4\left(\frac{1}{\log_5 x}\right) = -3$$

$$\log_5 x - \frac{4}{\log_5 x} = -3$$

Let $y = \log_5 x$.

Then, we have,

$$y - \frac{4}{y} = -3$$

$(\times y)$

$$y^2 - 4 = -3y$$

$$y^2 + 3y - 4 = 0$$

$$(y+4)(y-1) = 0$$

Either $y + 4 = 0$ or $y - 1 = 0$

$y = -4$ $y = 1$

Consider,

$$y = -4$$

$$\log_5 x = -4$$

Converting to exponential form gives:

$$x = 5^{-4}$$

$$x = \frac{1}{625}$$

Consider,

$$y = 1$$

$\log_5 x = 1$

Converting to exponential form gives:

$x = 5^1$

$x = 5$

$\therefore x = \frac{1}{625}$ and $x = 5$

(d) Prove that the function $f(x) = 3x - 2$ is bijective. [8]

In order to prove that the function $f(x) = 3x - 2$ is bijective, we need to prove that it is injective and surjective.

A function, f, is injective if $f(a) = f(b)$ implies $a = b$.

Since $f(x) = 3x - 2$,

Then $f(a) = 3a - 2$ and $f(b) = 3b - 2$.

If $f(a) = f(b)$

Then $3a - 2 = 3b - 2$

$\Rightarrow 3a = 3b$

$\Rightarrow \quad a = b$

Since $f(a) = f(b) \Rightarrow a = b$, then $f(x)$ is injective.

We now need to prove that $f(x)$ is surjective.

Let y be an element of the codomain.

$y = 3x - 2$

$y + 2 = 3x$

$x = \frac{y+2}{3}$

Since y is a real number, then $\frac{y+2}{3}$ is a real number.

So,

$$f\left(\frac{y+2}{3}\right) = 3\left(\frac{y+2}{3}\right) - 2$$

$$= y + 2 - 2$$

$$= y$$

Hence, for every x in the domain, there is a corresponding y in the codomain.

That is, $\forall y \in \mathbb{R}, \exists x \in \mathbb{R}$.

So, $f(x)$ is surjective.

$\therefore$ Since f is injective and surjective, f is bijective.

Q.E.D.

Total: 25 marks

June 2024 – Question 2 – Solution

(a) The roots of the cubic equation $x^3 - 9 = 0$ are α, β, and γ. Determine the cubic equation with roots $(\alpha - 1)$, $(\beta - 1)$ and $(\gamma - 1)$, given that

$(\alpha - 1)(\beta - 1) + (\alpha - 1)(\gamma - 1) + (\beta - 1)(\gamma - 1) = 12.$ [8]

$x^3 - 9 = 0$

where $a = 1, b = 0, c = 0$ and $d = -9$.

$$\alpha + \beta + \gamma = -\frac{b}{a}$$
$$= \frac{-(0)}{1}$$
$$= 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{0}{1}$$
$$= 0$$

$$\alpha\beta\gamma = -\frac{d}{a}$$
$$= -\frac{(-9)}{1}$$
$$= 9$$

For the new roots:

$$(\alpha - 1) + (\beta - 1) + (\gamma - 1) = (\alpha + \beta + \gamma) - 3$$
$$= 0 - 3$$
$$= -3$$

$$(\alpha-1)(\beta-1)+(\alpha-1)(\gamma-1)+(\beta-1)(\gamma-1)=12$$

$$\begin{aligned}(\alpha-1)(\beta-1)(\gamma-1) &= (\alpha\beta-\alpha-\beta+1)(\gamma-1)\\ &= \alpha\beta\gamma-\alpha\gamma-\beta\gamma+\gamma-\alpha\beta+\alpha+\beta-1\\ &= (\alpha+\beta+\gamma)-(\alpha\gamma+\alpha\beta+\beta\gamma)+(\alpha\beta\gamma)-1\\ &= 0-0+9-1\\ &= 8\end{aligned}$$

∴ The new equation is:

$$x^3-(sum\ of\ roots)x^2+(sum\ of\ the\ product\ of\ two\ of\ the\ roots\ at\ a\ time)x-(product\ of\ the\ roots)=0$$

$$x^3-(-3)x^2+(12)x-(8)=0$$

$$x^3+3x^2+12x-8=0$$

(b) The function $f(x) = 2x^3 - px^2 + qx - 5$ is divisible by $2x - 1$ and has a remainder of 7 when divided by $x - 1$. Calculate the values of p and q. [9]

$f(x) = 2x^3 - px^2 + qx - 5$

Since $f(x)$ is divisible by $(2x - 1)$, then by the Factor Theorem, $f\left(\frac{1}{2}\right) = 0$.

$f\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right)^3 - p\left(\frac{1}{2}\right)^2 + q\left(\frac{1}{2}\right) - 5$

$0 = 2\left(\frac{1}{8}\right) - p\left(\frac{1}{4}\right) + \frac{1}{2}q - 5$

$0 = \frac{1}{4} - \frac{1}{4}p + \frac{1}{2}q - 5$

$(\times 4)$

$0 = 1 - p + 2q - 20$

$p = 2q + 1 - 20$

$p = 2q - 19$ $\rightarrow$ Equation 1

Since $f(x)$ has a remainder of 7 when divided by $(x - 1)$, then by the Remainder Theorem, $f(1) = 7$.

$f(1) = 2(1)^3 - p(1)^2 + q(1) - 5$

$7 = 2 - p + q - 5$

$7 = -p + q - 3$

$p = q - 3 - 7$

$p = q - 10$ $\rightarrow$ Equation 2

Equating Equation 1 and Equation 2 gives:

$2q - 19 = q - 10$

$2q - q = 19 - 10$

$q = 9$

Substituting $q = 9$ into Equation 2 gives:

$p = (9) - 10$

$= -1$

$\therefore$ The values are $p = -1$ and $q = 9$.

(c) Let P be the proposition that if $n^3 + 3$ is odd then n is even for all integers n. Provide a proof by contradiction for this proposition.

Hint: Let $n^3 + 3$ be odd and assume that n is odd. Show that the assumption that n is odd is incorrect. [8]

Let P be the proposition that if $n^3 + 3$ is odd then n is even for all integers n.

Let $n^3 + 3$ be odd. Assume that n is odd.

Since n is assumed to be odd, then we can let $n = 2k + 1$, where $k \in \mathbb{Z}$.

$$\begin{aligned} n^3 + 3 &= (2k+1)^3 + 3 \\ &= (2k+1)^2(2k+1) + 3 \\ &= (4k^2 + 4k + 1)(2k+1) + 3 \\ &= 8k^3 + 8k^2 + 2k + 4k^2 + 4k + 1 + 3 \\ &= 8k^3 + 12k^2 + 6k + 4 \\ &= 2(4k^3 + 6k^2 + 3k + 2) \quad \text{which is even.} \end{aligned}$$

We assumed that $n^3 + 3$ was odd but we have shown it is even. This is a contradiction. The assumption that n is odd must be false. So, n must be even.

$\therefore$ We have proven by contradiction that if $n^3 + 3$ is odd then n is even for all integers n.

Q.E.D.

Total: 25 marks

June 2024 – Question 3 – Solution

(a) (i) Express $5\sin\theta + 12\cos\theta$ in the form $r\sin(\theta + \alpha)$, where $r > 0$ and $0 < \alpha < 2\pi$. [6]

Let $5\sin\theta + 12\cos\theta = r\sin(\theta + \alpha)$

$$5\sin\theta + 12\cos\theta = r\sin\theta\cos\alpha + r\cos\theta\sin\alpha$$

Equating coefficients gives:

$r\cos\alpha = 5$ → Equation 1

$r\sin\alpha = 12$ → Equation 2

Equation 2 ÷ Equation 1 gives,

$$\frac{r\sin\alpha}{r\cos\alpha} = \frac{12}{5}$$

$$\tan\alpha = \frac{12}{5}$$

$$\alpha = \tan^{-1}\left(\frac{12}{5}\right)$$

$$\alpha = 1.18^c \quad \text{(to 3 significant figures)}$$

$$r = \sqrt{(5)^2 + (12)^2}$$

$$= \sqrt{25 + 144}$$

$$= \sqrt{169}$$

$$= 13$$

$$\therefore 5\sin\theta + 12\cos\theta = 13\sin(\theta + 1.18^c)$$

(ii) Hence, or otherwise, show that $5\sin\theta + 12\cos\theta + 7 \leq 20$. [3]

$$5\sin\theta + 12\cos\theta = 13\sin(\theta + 1.18^c)$$

$$5\sin\theta + 12\cos\theta + 7 = 13\sin(\theta + 1.18^c) + 7$$

Now,

$$-1 \leq \sin\theta \leq 1$$

$$-1 \leq \sin(\theta + 1.18^c) \leq 1$$

$$-13 \leq 13\sin(\theta + 1.18^c) \leq 13$$

$$-13 + 7 \leq 13\sin(\theta + 1.18^c) + 7 \leq 13 + 7$$

$$-6 \leq 13\sin(\theta + 1.18^c) + 7 \leq 20$$

$$-6 \leq 5\sin\theta + 12\cos\theta + 7 \leq 20$$

$$\therefore 5\sin\theta + 12\cos\theta + 7 \leq 20$$

Q.E.D.

(b) (i) Show that $2\sin x + 3\cos x = 3$ may be written as

$13\cos^2 x - 18\cos x + 5 = 0.$ [5]

$2\sin x + 3\cos x = 3$

$2\sin x = 3 - 3\cos x$

Squaring both sides gives:

$(2\sin x)^2 = (3 - 3\cos x)^2$

$4\sin^2 x = 9 - 18\cos x + 9\cos^2 x$

$4(1 - \cos^2 x) = 9 - 18\cos x + 9\cos^2 x$

$4 - 4\cos^2 x = 9 - 18\cos x + 9\cos^2 x$

$0 = 9\cos^2 x + 4\cos^2 x - 18\cos x + 9 - 4$

$0 = 13\cos^2 x - 18\cos x + 5$

$\therefore \sin x + 3\cos x = 3$ may be written as $13\cos^2 x - 18\cos x + 5 = 0$

Q.E.D.

(ii) Hence, or otherwise, solve the equation $2\sin x + 3\cos x = 3$ for $-\pi \le x \le \pi$. [5]

$$2\sin x + 3\cos x = 3$$

$$13\cos^2 x - 18\cos x + 5 = 0$$

$$(13\cos x - 5)(\cos x - 1) = 0$$

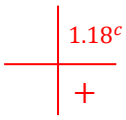
Either $\quad 13\cos x - 5 = 0 \quad$ or $\quad \cos x - 1 = 0$

$$13\cos x = 5 \qquad \cos x = 1$$

$$\cos x = \frac{5}{13}$$

Consider,

$\cos x = \frac{5}{13}$



$x = -1.18^c, 1.18^c$

Consider,

$\cos x = 1$

$x = 0$

$\therefore x = -1.18^c, 0, 1.18^c$ for $-\pi \le x \le \pi$.

(c) Show that $\frac{(2\cos^2 x-1)^2}{\cos^4 x-\sin^4 x} \equiv 1-2\sin^2 x$. [6]

Required to show that $\frac{(2\cos^2 x-1)^2}{\cos^4 x-\sin^4 x} \equiv 1-2\sin^2 x$.

Taking L.H.S:

$$\frac{(2\cos^2 x-1)^2}{\cos^4 x-\sin^4 x} = \frac{(\cos 2x)^2}{(\cos^2 x-\sin^2 x)(\cos^2 x+\sin^2 x)}$$

$$= \frac{(\cos 2x)(\cos 2x)}{(\cos 2x)(\cos^2 x+\sin^2 x)}$$

$$= \frac{\cos 2x}{\cos^2 x+\sin^2 x}$$

$$= \frac{1-2\sin^2 x}{(1-\sin^2 x)+\sin^2 x}$$

$$= \frac{1-2\sin^2 x}{1}$$

$$= 1-2\sin^2 x$$

$$= \text{R.H.S.}$$

$$\therefore \frac{(2\cos^2 x-1)^2}{\cos^4 x-\sin^4 x} = 1-2\sin^2 x$$

Q.E.D.

Total: 25 marks

June 2024 – Question 4 – Solution

(a) A plane, P, passes through the point $(5, 1, -3)$ and is perpendicular to the vector

$n = 3\hat{\boldsymbol{\imath}} - 4\hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}$.

(i) Show that the equation of the plane is $3x - 4y + 2z - 5 = 0$. [4]

Let $\boldsymbol{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $\boldsymbol{a} = \begin{pmatrix} 5 \\ 1 \\ -3 \end{pmatrix}$ and $\boldsymbol{n} = \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix}$.

The Cartesian equation of the plane is:

$$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix}$$

$$(x)(3) + (y)(-4) + (z)(2) = (5)(3) + (1)(-4) + (-3)(2)$$

$$3x - 4y + 2z = 15 - 4 - 6$$

$$3x - 4y + 2z = 5$$

$$3x - 4y + 2z - 5 = 0$$

$\therefore$ The equation of the plane is $3x - 4y + 2z - 5 = 0$.

Q.E.D.

(ii) Calculate the angle, in radians, between P and the plane with equation

$7x - 4y + 3z = 5$. [6]

We have two equations:

$3x - 4y + 2z - 5 = 0$

$7x - 4y + 3z = 5$

Let $\boldsymbol{a} = \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} 7 \\ -4 \\ 3 \end{pmatrix}$.

$$\boldsymbol{a} \cdot \boldsymbol{b} = \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ -4 \\ 3 \end{pmatrix}$$

$$= (3)(7) + (-4)(-4) + (2)(3)$$

$$= 21 + 16 + 6$$

$$= 43$$

$$|\boldsymbol{a}| = \sqrt{(3)^2 + (-4)^2 + (2)^2}$$

$$= \sqrt{9 + 16 + 4}$$

$$= \sqrt{29}$$

$$|\boldsymbol{b}| = \sqrt{(7)^2 + (-4)^2 + (3)^2}$$

$$= \sqrt{49 + 16 + 9}$$

$$= \sqrt{74}$$

Now,

$$\cos\theta = \frac{\boldsymbol{a}\cdot\boldsymbol{b}}{|\boldsymbol{a}||\boldsymbol{b}|}$$

$$\cos\theta = \frac{43}{\sqrt{29}\sqrt{74}}$$

$$\theta = \cos^{-1}\left(\frac{43}{\sqrt{29}\sqrt{74}}\right)$$

$$\theta = 0.381^c \quad \text{(to 3 significant figures)}$$

$\therefore$ The angle is $\theta = 0.381^c$.

(b) The equation of a circle is given as $x^2 + y^2 - 6x - 8y - 39 = 0$.

(i) Determine the centre and radius of the circle. [5]

$$x^2 + y^2 - 6x - 8y - 39 = 0$$

$$x^2 - 6x + y^2 - 8y = 39$$

$$x^2 - 6x + 9 + y^2 - 8y + 16 = 39 + 9 + 16$$

$$(x-3)^2 + (y-4)^2 = 64$$

$$(x-3)^2 + (y-4)^2 = 8^2$$

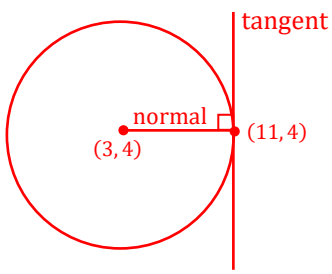
which is in the form $(x-a)^2 + (y-b)^2 = r^2$,

where the centre is $C(3,4)$ and $r = 8$ units.

$\therefore$ The centre of the circle is $C(3,4)$ and the radius of the circle is $r = 8$ units.

(ii) Determine the equation of the tangent to the circle at the point $(11,4)$. [5]

Consider the diagram below:



The normal to the circle has points $(3,4)$ and $(11,4)$.

So, we have,

Gradient of normal $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{4-4}{11-3}$

$= \frac{0}{8}$

$= 0$

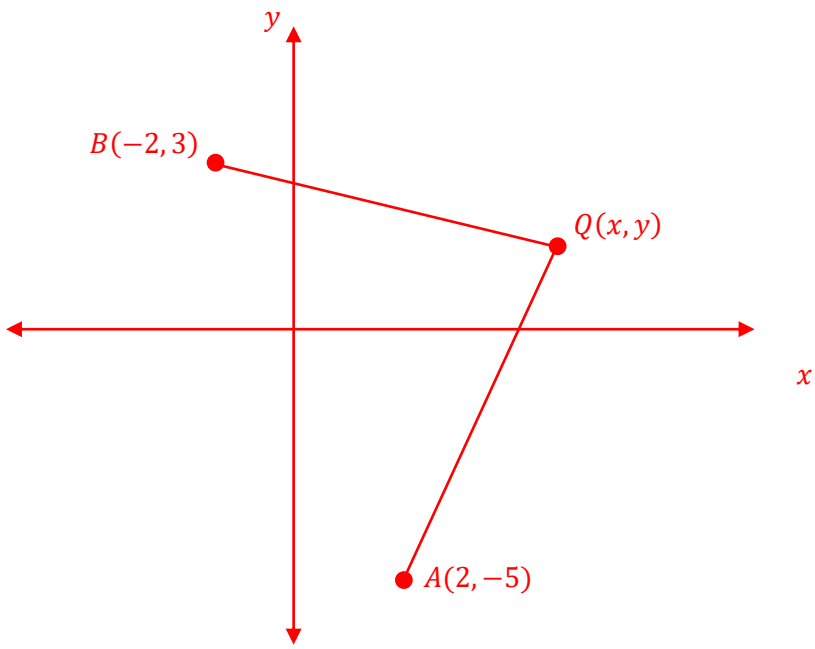
Since the gradient of the normal is equal to zero, then the normal is a horizontal line.

So, the tangent is a vertical line that passes through the point $(11, 4)$.

$\therefore$ The equation of the tangent that touches the circle at $(11, 4)$ is: $x = 11$

(c) A point, Q, moves in the xy-plane such that it is equidistant from $A(2,-5)$ and $B(-2,3)$. Determine the locus of Q. [5]

Consider the diagram below:



Consider points $A(2,-5)$ and $Q(x,y)$.

Distance $= \sqrt{(x-2)^2+(y+5)^2}$

Consider points $B(-2,3)$ and $Q(x,y)$.

Distance $= \sqrt{(x+2)^2+(y-3)^2}$

Now,

$$\sqrt{(x-2)^2+(y+5)^2} = \sqrt{(x+2)^2+(y-3)^2}$$

Squaring both sides gives,

$$(x-2)^2+(y+5)^2=(x+2)^2+(y-3)^2$$

$$x^2-4x+4+y^2+10y+25=x^2+4x+4+y^2-6y+9$$

$$-4x+10y+25=4x-6y+9$$

$$10y+6y=4x+4x+9-25$$

$$16y=8x-16$$

$$(\div 8)$$

$$2y=x-2$$

$\therefore$ The equation of the locus of Q is: $2y=x-2$

Total: 25 marks

June 2024 – Question 5 – Solution

(a) A function f is defined as $f(x) = \begin{cases} (x+4)^3 & , x < -2 \\ -10 & , x = -2 \\ x^2 + 8x - 1 & , x > -2 \end{cases}$

(i) Determine $f(-2)$. [1]

From definition of function,

$f(-2) = -10$

(ii) Determine $\lim_{x \to -2} f(x)$. [5]

$$\lim_{x \to -2^-} f(x) = \lim_{x \to -2^-} (x+4)^3$$

$$= (-2+4)^3$$

$$= (2)^3$$

$$= 8$$

$$\lim_{x \to -2^+} f(x) = \lim_{x \to -2^+} (x^2 + 8x - 1)$$

$$= (-2)^2 + 8(-2) - 1$$

$$= 4 - 16 - 1$$

$$= -13$$

Since $\lim_{x \to -2^-} f(x) = 8$ and $\lim_{x \to -2^+} f(x) = -13$, then $\lim_{x \to -2^-} f(x) \neq \lim_{x \to -2^+} f(x)$.

$\therefore$ The $\lim_{x \to -2} f(x)$ does not exist.

(iii) Hence, or otherwise, determine whether f is continuous at $x = -2$. [1]

Since $\lim_{x \to -2} f(x)$ does not exist, then f is not continuous at $x = -2$.

(b) Evaluate $\lim_{\theta \to 0} \left(\frac{\sin \theta}{\sin 4\theta}\right)$. [4]

$$\lim_{\theta \to 0} \left(\frac{\sin \theta}{\sin 4\theta}\right) = \lim_{\theta \to 0} \left(\frac{\sin \theta}{\theta} \times \frac{\theta}{\sin 4\theta}\right)$$

$$= \lim_{\theta \to 0} \left(\frac{\sin \theta}{\theta} \div \frac{\sin 4\theta}{\theta}\right)$$

$$= \lim_{\theta \to 0} \left(\frac{\sin \theta}{\theta}\right) \div \lim_{\theta \to 0} \left(\frac{\sin 4\theta}{\theta}\right)$$

$$= 1 \div \lim_{\theta \to 0} \left(\frac{\sin 4\theta}{\theta} \times \frac{4}{4}\right) \qquad \left[\because \lim_{\theta \to 0} \left(\frac{\sin \theta}{\theta}\right) = 1\right]$$

$$= 1 \div 4 \lim_{\theta \to 0} \left(\frac{\sin 4\theta}{4\theta}\right)$$

$$= 1 \div 4(1) \qquad \left[\because \lim_{\theta \to 0} \left(\frac{\sin 4\theta}{4\theta}\right) = 1\right]$$

$$= 1 \div 4$$

$$= \frac{1}{4}$$

$$\therefore \lim_{\theta \to 0} \left(\frac{\sin \theta}{\sin 4\theta}\right) = \frac{1}{4}$$

(c) A function, g, defined as $g(x) = x^3 - 3x^2$.

(i) Determine the stationary points of g and the nature of the stationary points. [9]

$g(x) = x^3 - 3x^2$

$g'(x) = 3x^2 - 6x$

At stationary points, $g'(x) = 0$.

When $g'(x) = 0$,

$3x^2 - 6x = 0$

$3x(x - 2) = 0$

Either $3x = 0$ or $x - 2 = 0$

$x = 0$ $x = 2$

When $x = 0$,

$g(0) = (0)^3 - 3(0)^2$

$= 0 - 0$

$= 0$

When $x = 2$,

$g(2) = (2)^3 - 3(2)^2$

$= 8 - 12$

$= -4$

$\therefore g(x)$ has stationary points at $(0, 0)$ and $(2, -4)$.

$g'(x) = 3x^2 - 6x$

$g''(x) = 6x - 6$

When $x = 0$,

$g''(0) = 6(0) - 6$

$= 0 - 6$

$= -6 \quad (< 0)$

Since $g''(0) < 0$, then $(0, 0)$ is a maximum point.

When $x = 2$,

$g''(2) = 6(2) - 6$

$= 12 - 6$

$= 6 \quad (> 0)$

Since $g''(2) > 0$, then $(2, -4)$ is a minimum point.

(ii) Hence, or otherwise, sketch the graph of g showing its intercepts, stationary points and any other important features. [5]

$g(x) = x^3 - 3x^2$

The y-intercept occurs at $(0, 0)$.

When $g(x) = 0$,

$0 = x^3 - 3x^2$

$0 = x^2(x - 3)$

Either $x^2 = 0$ or $x - 3 = 0$

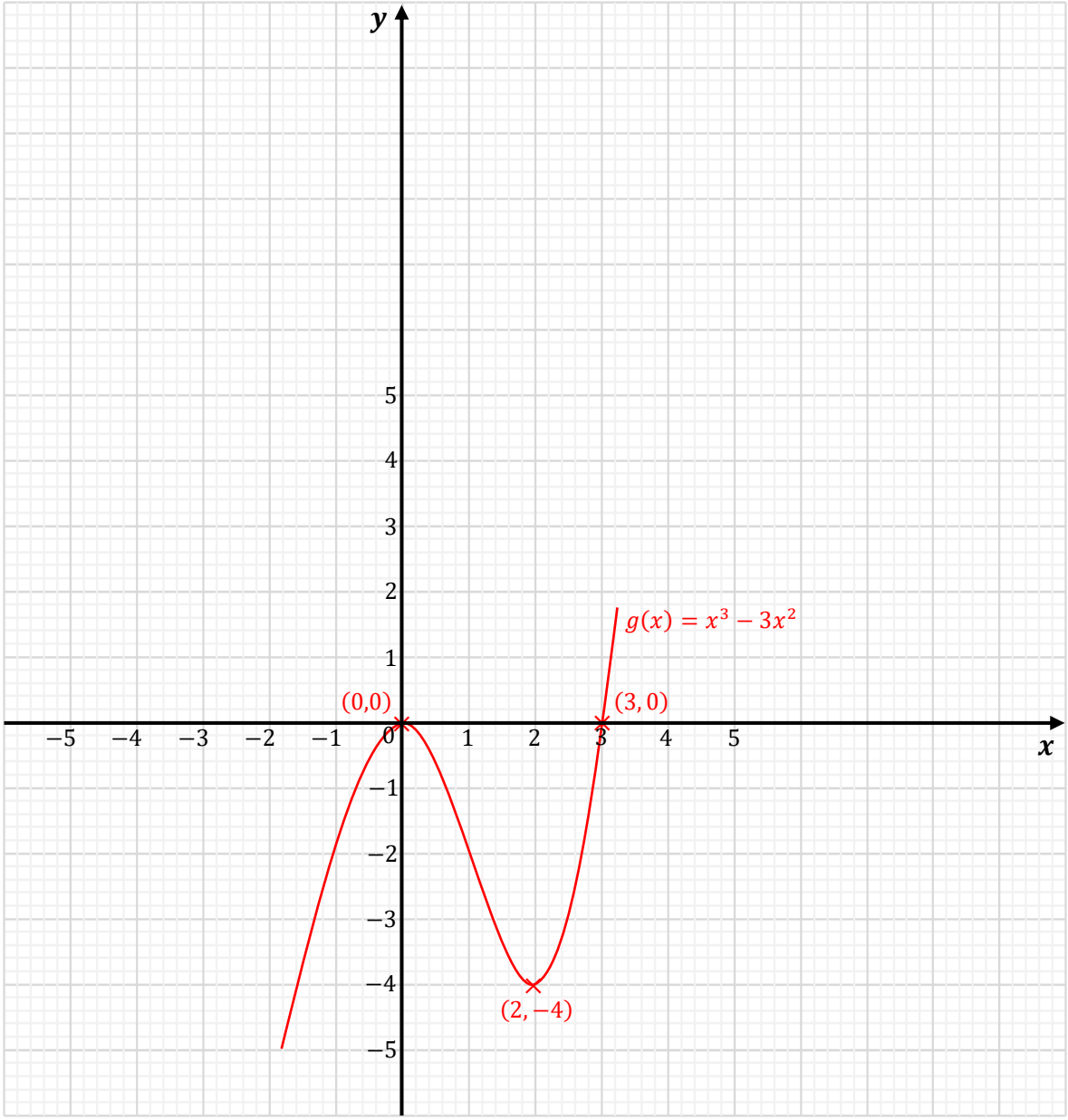
$x = 0$ $x = 3$

The x-intercepts are $(0, 0)$ and $(3, 0)$.

The stationary points are $(0, 0)$ and $(2, -4)$.

$(0, 0)$ is a maximum point and $(2, -4)$ is a minimum point.

The sketch of $g(x) = x^3 - 3x^2$ is shown below:



Total: 25 marks

June 2024 – Question 6 – Solution

(a) Calculate the volume of the solid generated by revolving the region bounded by the graphs of $x = \sqrt{y}$ and $y = 2x$ about the y-axis. [7]

$x = \sqrt{y}$

Squaring both sides gives:

$x^2 = y$

$y = x^2 \quad \rightarrow$ Equation 1

$y = 2x \quad \rightarrow$ Equation 2

Equating both equations gives:

$$x^2 = 2x$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

Either $\quad x = 0 \quad$ or $\quad x - 2 = 0$

$$x = 2$$

When $x = 0$,

$y = 2(0)$

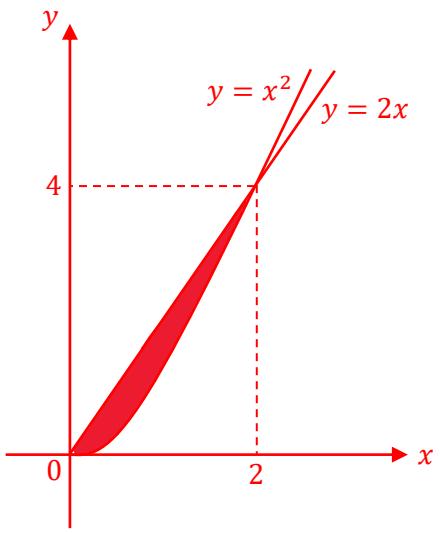
$y = 0$

When $x = 2$,

$y = 2(2)$

$y = 4$

Consider the sketch below of $y = x^2$ and $y = 2x$.



Let Curve 1 be $y = x^2$ and let Curve 2 be $y = 2x$.

Consider,

$y = x^2$

$$\text{Volume about Curve 1} = \pi \int_a^b x^2\, dy$$

$$= \pi \int_0^4 y\, dy$$

$$= \pi \left[\frac{y^2}{2}\right]_0^4$$

$$= \pi \left[\frac{(4)^2}{2} - \frac{(0)^2}{2}\right]$$

$$= \pi \left(\frac{16}{2} - \frac{0}{2}\right)$$

$$= \pi(8 - 0)$$

$$= 8\pi \text{ units}^3$$

Now,

$$y = 2x$$

$$\frac{y}{2} = x$$

$$\left(\frac{y}{2}\right)^2 = x^2$$

$$\frac{y^2}{4} = x^2$$

Volume about Curve 2 $= \pi \int_a^b x^2\, dy$

$$= \pi \int_0^4 \left(\frac{y^2}{4}\right) dy$$

$$= \frac{\pi}{4} \int_0^4 (y^2)\, dy$$

$$= \frac{\pi}{4} \left[\frac{y^3}{3}\right]_0^4$$

$$= \frac{\pi}{12} [(4)^3 - (0)^3]$$

$$= \frac{\pi}{12} (64 - 0)$$

$$= \frac{\pi}{12} (64)$$

$$= \frac{16\pi}{3} \text{ units}^3$$

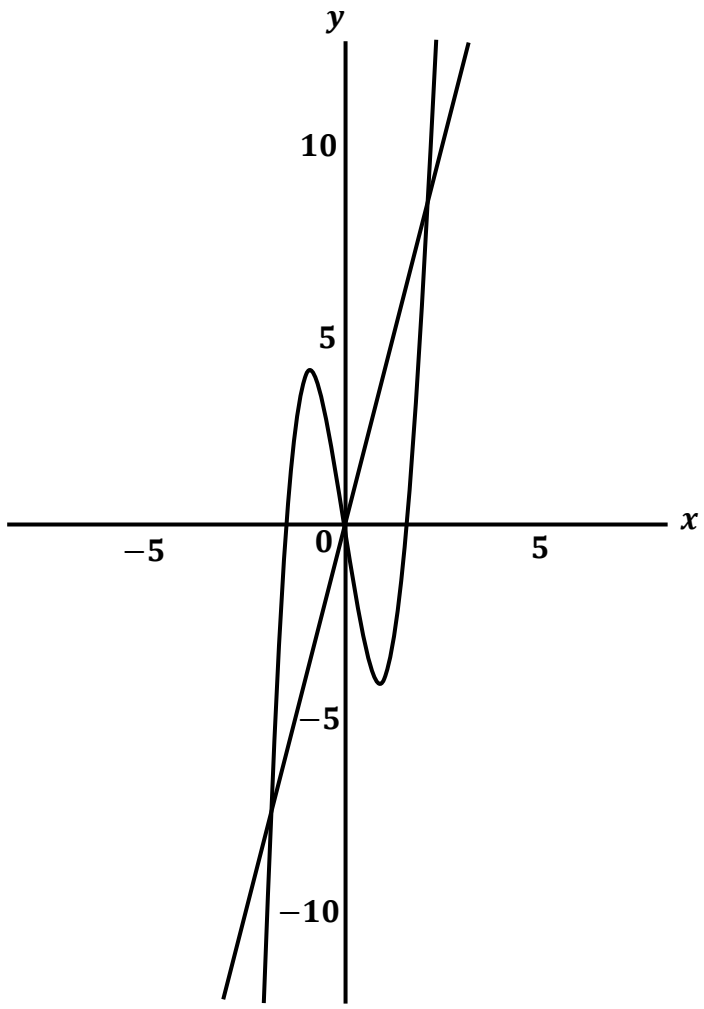
Hence,

Volume of solid about y-axis = Volume about Curve 1 − Volume about Curve 2

$$= 8\pi - \frac{16\pi}{3}$$

$$= \frac{8\pi}{3} \text{ units}^3$$

(b) The diagram below, not drawn to scale, shows the graphs of $y = 2x^3 - 6x$ and $y = 4x$.



Calculate the area between the line and the curve. [7]

To calculate the area between the line and the curve, we need to find the points of intersection.

$y = 2x^3 - 6x$ → Equation 1

$y = 4x$ → Equation 2

Equating both equations gives:

$$2x^3 - 6x = 4x$$

$$2x^3 - 6x - 4x = 0$$

$$2x^3 - 10x = 0$$

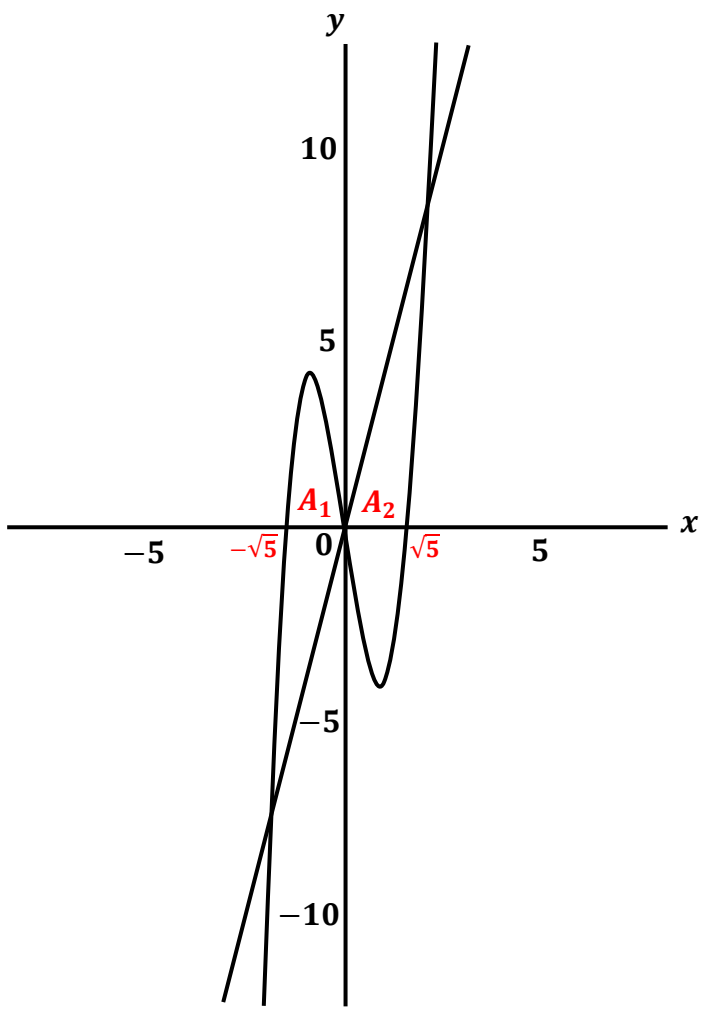
$$2x(x^2 - 5) = 0$$

$$2x(x - \sqrt{5})(x + \sqrt{5}) = 0$$

Either $2x = 0$ or $x - \sqrt{5} = 0$ or $x + \sqrt{5} = 0$

$x = 0$ $x = \sqrt{5}$ $x = -\sqrt{5}$

Consider the sketch below:



Area of A_1 = Area under curve − Area under line

$$= \int_{-\sqrt{5}}^{0} (2x^3 - 6x)\, dx - \int_{-\sqrt{5}}^{0} (4x)\, dx$$

$$= \int_{-\sqrt{5}}^{0} (2x^3 - 6x - 4x)\, dx$$

$$= \int_{-\sqrt{5}}^{0} (2x^3 - 10x)\, dx$$

$$= \left[\frac{x^4}{2} - 5x^2\right]_{-\sqrt{5}}^{0}$$

$$= \left[\frac{(0)^4}{2} - 5(0)^2\right] - \left[\frac{(-\sqrt{5})^4}{2} - 5(-\sqrt{5})^2\right]$$

$$= \left(\frac{0}{2} - 0\right) - \left[\frac{25}{2} - 5(5)\right]$$

$$= 0 - \left(\frac{25}{2} - 25\right)$$

$$= 0 - \left(-\frac{25}{2}\right)$$

$$= \frac{25}{2} \text{ units}^2$$

Area of A_2 = Area under line − Area under curve

$$= \int_{0}^{\sqrt{5}} (4x)\, dx - \int_{0}^{\sqrt{5}} (2x^3 - 6x)\, dx$$

$$= \int_{0}^{\sqrt{5}} (4x - 2x^3 + 6x)\, dx$$

$$= \int_{0}^{\sqrt{5}} (10x - 2x^3)\, dx$$

$$= \left[5x^2 - \frac{x^4}{2}\right]_{0}^{\sqrt{5}}$$

$$= \left[5(\sqrt{5})^2 - \frac{(\sqrt{5})^4}{2}\right] - \left[5(0)^2 - \frac{(0)^4}{2}\right]$$

$$= \left[5(5) - \frac{25}{2}\right] - 0$$

$$= 25 - \frac{25}{2}$$

$$= \frac{25}{2} \text{ units}^2$$

Hence,

$$\text{Area between line and curve} = \text{Area of } A_1 + \text{Area of } A_2$$

$$= \frac{25}{2} + \frac{25}{2}$$

$$= \frac{50}{2}$$

$$= 25 \text{ units}^2$$

(c) Given that f is an **even** function, show that $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$. [5]

Recall: A function $f(x)$ is said to be an even function if and only if $f(-x) = f(x)$.

Required to show that $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$.

Taking L.H.S:

$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$

Consider $\int_{-a}^{0} f(x)\,dx$.

Since $f(x)$ is even, then $f(-x) = f(x)$.

Let $u = -x$

$\frac{du}{dx} = -1$

$dx = -du$

Also, $x = -u$.

When $x = -a$,

$u = -(-a)$

$= a$

When $x = 0$,

$u = -0$

$= 0$

So, we have,

$$\int_{-a}^{0} f(x)\,dx = \int_{a}^{0} f(-u)\,(-du)$$

$$= -\int_{a}^{0} f(-u)\,du$$

$$= \int_{0}^{a} f(-u)\,du$$

$$= \int_{0}^{a} f(u)\,du \qquad [\because f \text{ is even}]$$

$$= \int_{0}^{a} f(x)\,dx$$

Now,

$$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$$

$$= \int_{0}^{a} f(x)\,dx + \int_{0}^{a} f(x)\,dx$$

$$= 2\int_{0}^{a} f(x)\,dx$$

$$\therefore \int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

Q.E.D.

(d) Solve the differential equation $x^3y' = 2 - x^4$, given that at $x = 1, y = 2$. [6]

$$x^3y' = 2 - x^4$$

$$x^3 \cdot \frac{dy}{dx} = 2 - x^4$$

$$\frac{dy}{dx} = \frac{2-x^4}{x^3}$$

$$\frac{dy}{dx} = 2x^{-3} - x$$

Now,

$$y = \int \frac{dy}{dx}\, dx$$

$$= \int (2x^{-3} - x)\, dx$$

$$= \frac{2x^{-2}}{-2} - \frac{x^2}{2} + c$$

$$= -\frac{1}{x^2} - \frac{x^2}{2} + c \qquad \rightarrow \text{G.S.}$$

Substituting $x = 1$ and $y = 2$ into $y = -\frac{1}{x^2} - \frac{x^2}{2} + c$ gives:

$$(2) = -\frac{1}{(1)^2} - \frac{(1)^2}{2} + c$$

$$2 = -1 - \frac{1}{2} + c$$

$$2 + 1 + \frac{1}{2} = c$$

$$\frac{7}{2} = c$$

$\therefore$ The particular solution is: $y = -\frac{1}{x^2} - \frac{x^2}{2} + \frac{7}{2}$

Total: 25 marks